



Experiment 2

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Subject Name: Full Stack Development-II

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1. Aim:

To implement Single Page Application (SPA) navigation in the EcoTrack application using React Router, secure application routes using context-based authentication, and extend nested dashboard routing through follow-up enhancements

2. Objective:

- To implement seamless Single Page Application (SPA) navigation in EcoTrack using React Router.
- To secure application routes through context-based authentication and protected routing.
- To design nested dashboard routes for modular and scalable UI navigation.
- To manage user session state globally using React Context API.
- To enhance dashboard functionality with future-ready, extensible routing architecture.

3. Implementation/Code:

dashboard.jsx:

```
dashboard.jsx: import { logs } from '../data/logs'; const Dashboard = () => { const  
totalcarbon = logs.reduce((total, log) => total + log.carbon, 0); return (
```

Dashboard Total Carbon Footprint: {totalcarbon} kg CO₂

```
{logs.map((log) => ( {log.activity} {log.carbon} kg CO2  
))}  
); };
```

```
export default Dashboard;
```

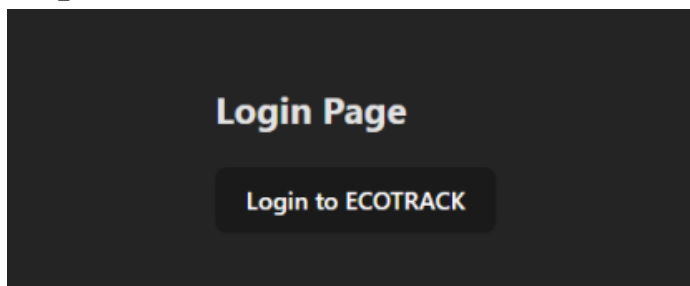
Logs.jsx

```
import {useAuth} from '../context/AuthContext';
import {useNavigate} from 'react-router-dom';
const LoginPage=()=>useAuth();
const {setIsAuthenticated}=useAuth();
const navigate=useNavigate();
const handleLogin=()=>{
    // Perform authentication logic here (e.g., API call)
    // On successful authentication: setIsAuthenticated(true);
    navigate('/'); // Redirect to a protected route after login
}
return(
<>
  <div>
    <h2> Login Page </h2>
    <button onClick={handleLogin}>Login to ECOTRACK</button>
  </div>
</>
);} export default handleLogin;
```

ProtectedRoute.jsx:

```
import { Navigate } from "react-router-dom";
import { useAuth } from
"../context/AuthContext";
const ProtectedRoute = ({ children }) => {
  const { isAuthenticated } = useAuth();
  return isAuthenticated ? children : <Navigate to="/login" />;
};
export default ProtectedRoute;
```

4. Output:





5. Learning Outcome:

- Understand how to build Single Page Applications using React Router for smooth client-side navigation.
- Learn to implement protected routes using Context API-based authentication.
- Gain hands-on experience with nested routing for structured dashboard layouts.
- Develop skills in managing global user state and session handling in React.
- Learn to design scalable and maintainable routing architectures for realworld applications.